DATE: **EXERCISE NO:** **EXERCISE NAME:**

WEEK-1: HTML Frames: Frame Structure: Design a webpage for the online book store using HTML frames. Include a navigation frame, a content frame for displaying book details, and a footer frame for additional information.

WEEK1.html:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Online Book Store</title>
  </head>
  <frameset rows="20%,70%,10%" border="2">
    <iframe src=""></iframe>
    <frame src="navigation.html" name="navFrame">
    <frame src="content.html" name="contentFrame">
    <frame src="footer.html" name="footerFrame">
  </frameset>
</html>
```

Navigation.html:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Navigation</title>
  </head>
  <body bgcolor="BBBBBB">
    <h3>Book Categories</h3>
    <h1>Hello World Welcome to VNRVJIET</h1>
    <ul>
      <li><a href="https://www.microsoft.com" target="contentFrame">Computer Science</a></li>
      <li><a href="https://www.ieee.org target="contentFrame">Electronics</a></li>
      <li><a href="content.html" target="contentFrame">Mechanical</a></li>
      <li><a href="content.html" target="contentFrame">Civil</a></li>
    </ul>
  </body>
</html>
```

Footer.html:

```
<!DOCTYPE html>
<head>
  <title>Footer</title>
</head>
<body bgcolor="90EE90" align="center">
  <p>2025 Online Book Store | Contact: Support@bookstore.com</p>
</body>
</html>
```

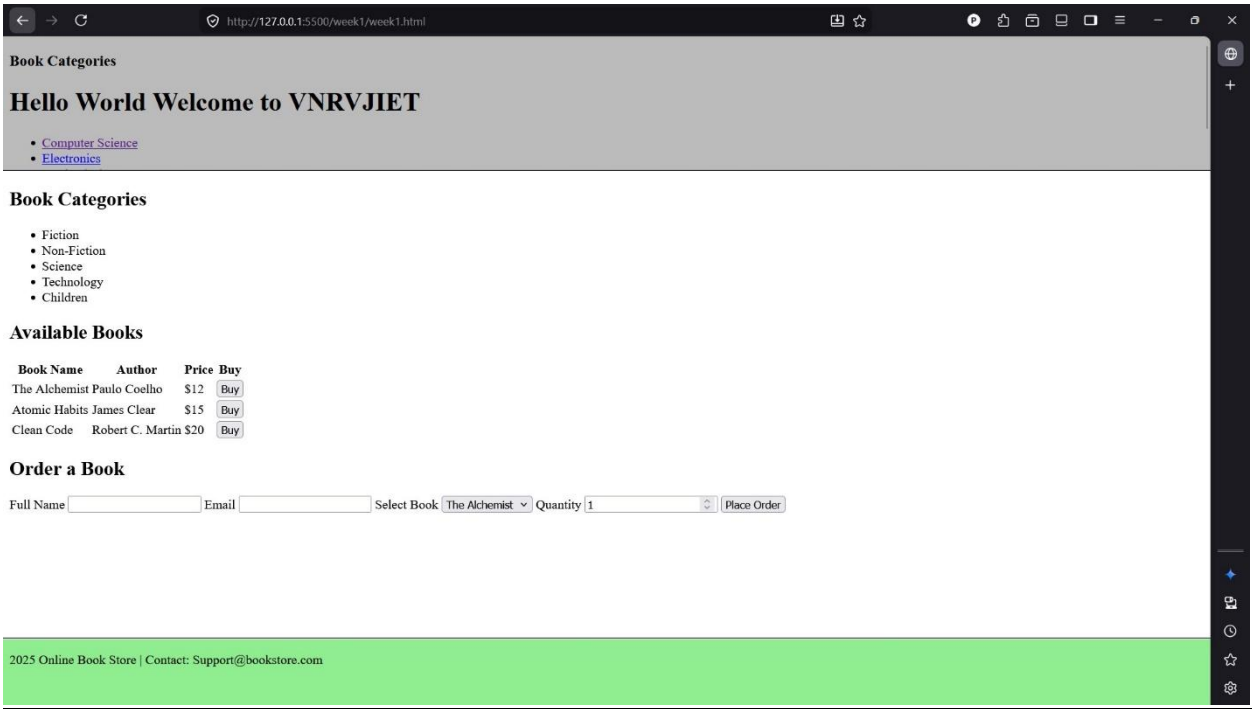
Content.html:

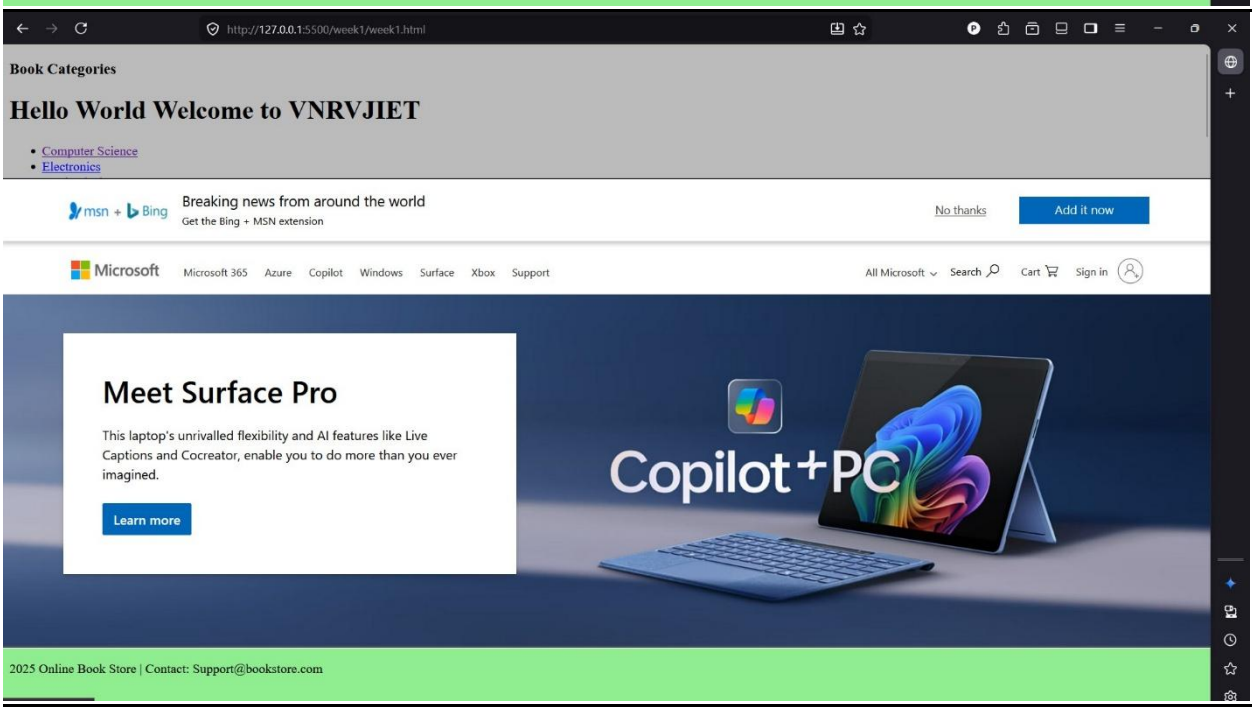
```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Book Store</title>
   <link rel="stylesheet" href="content.css">
</head>
<body>

   <section class="categories">
      <h2>Book Categories</h2>
      <ul>
         <li>Fiction</li>
         <li>Non-Fiction</li>
         <li>Science</li>
         <li>Technology</li>
         <li>Children</li>
      </ul>
   </section>

   <section class="books">
      <h2>Available Books</h2>
      <table>
         <tr>
            <th>Book Name</th>
            <th>Author</th>
            <th>Price</th>
            <th>Buy</th>
         </tr>
         <tr>
            <td>The Alchemist</td>
            <td>Paulo Coelho</td>
            <td>$12</td>
            <td><button>Buy</button></td>
         </tr>
         <tr>
            <td>Atomic Habits</td>
            <td>James Clear</td>
            <td>$15</td>
            <td><button>Buy</button></td>
         </tr>
         <tr>
            <td>Clean Code</td>
            <td>Robert C. Martin</td>
            <td>$20</td>
            <td><button>Buy</button></td>
         </tr>
```

```
      </table>
   </section>

   <section class="order">
      <h2>Order a Book</h2>
      <form>
         <label>Full Name</label>
         <input type="text" required>

         <label>Email</label>
         <input type="email" required>

         <label>Select Book</label>
         <select>
            <option>The Alchemist</option>
            <option>Atomic Habits</option>
            <option>Clean Code</option>
         </select>

         <label>Quantity</label>
         <input type="number" min="1" value="1">

         <button type="submit">Place Order</button>
      </form>
   </section>

</body>
</html>
```

WEBPAGE:

DATE: **EXERCISE NO:** **EXERCISE NAME:**





DATE: **EXERCISE NO:** **EXERCISE NAME:**

WEEK-2: CSS Styling: Apply CSS styling to enhance the visual appeal of the frames and content. Use appropriate styles for fonts, colors, and layout.

Week2.html:

```
<!DOCTYPE html>
<html>
   <head>
      <title>Online Book Store</title>
   </head>
   <frameset rows="20%,70%,10%" border="2">
      <iframe src=""></iframe>
      <frame src="navigation.html" name="navFrame">
      <frame src="content.html" name="contentFrame">
      <frame src="footer.html" name="footerFrame">
   </frameset>
</html>
```

Navigation.html:

```
<!DOCTYPE html>
<html>
   <head>
      <title>Navigation</title>
   </head>
   <body bgcolor="BBBBBB">
      <h3>Book Categories</h3>
      <h1>Hello World Welcome to VNRVJIET</h1>
      <ul>
         <li><a href="https://www.google.com" target="contentFrame">Computer Science</a></li>
         <li><a href="content.html" target="contentFrame">Electronics</a></li>
         <li><a href="content.html" target="contentFrame">Mechanical</a></li>
         <li><a href="content.html" target="contentFrame">Civil</a></li>
      </ul>
   </body>
</html>
```

Navigation.css:

```
body {
   font-family: Arial, sans-serif;
   background-color: #eeeeee;
   padding: 10px;
}

h3 {
   color: #2c3e50;
}

ul {
   padding-left: 15px;
```

```
}

li {
   margin: 6px 0;
}

a {
   text-decoration: none;
   color: #2980b9;
}

a:hover {
   text-decoration: underline;
}
```

Footer.html:

```
<!DOCTYPE html>
<head>
   <title>Footer</title>
</head>
<body bgcolor="90EE90" align="center">
   <p>2025 Online Book Store | Contact: Support@bookstore.com</p>
</body>
</html>
```

Footer.css:

```
body {
   font-family: Arial, sans-serif;
   background-color: #90EE90;
   margin: 0;
   text-align: center;
}

p {
   font-size: 14px;
   padding-top: 10px;
}
```

Content.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Book Store</title>
   <link rel="stylesheet" href="content.css">
</head>
<body>

   <!-- Categories List -->
```

```
<section class="categories">
  <h2>Book Categories</h2>
  <ul>
    <li>Fiction</li>
    <li>Non-Fiction</li>
    <li>Science</li>
    <li>Technology</li>
    <li>Children</li>
  </ul>
</section>

<!-- Books Table -->
<section class="books">
  <h2>Available Books</h2>
  <table>
    <tr>
      <th>Book Name</th>
      <th>Author</th>
      <th>Price</th>
      <th>Buy</th>
    </tr>
    <tr>
      <td>The Alchemist</td>
      <td>Paulo Coelho</td>
      <td>$12</td>
      <td><button>Buy</button></td>
    </tr>
    <tr>
      <td>Atomic Habits</td>
      <td>James Clear</td>
      <td>$15</td>
      <td><button>Buy</button></td>
    </tr>
    <tr>
      <td>Clean Code</td>
      <td>Robert C. Martin</td>
      <td>$20</td>
      <td><button>Buy</button></td>
    </tr>
  </table>
</section>

<!-- Order Form -->
<section class="order">
  <h2>Order a Book</h2>
  <form>
    <label>Full Name</label>
    <input type="text" required>
```

```
        <label>Email</label>
        <input type="email" required>

        <label>Select Book</label>
        <select>
          <option>The Alchemist</option>
          <option>Atomic Habits</option>
          <option>Clean Code</option>
        </select>

        <label>Quantity</label>
        <input type="number" min="1" value="1">

        <button type="submit">Place Order</button>
      </form>
    </section>

</body>
</html>
```

Content.css:

```
/* General page styling */
body {
  font-family: Arial, sans-serif;
  background-color: #f5f7fa;
  margin: 0;
  padding: 15px;
  color: #333;
}

/* Section styling */
section {
  background: #ffffff;
  padding: 15px;
  margin-bottom: 20px;
  border-radius: 6px;
  box-shadow: 0 2px 6px rgba(0,0,0,0.1);
}

/* Headings */
h2 {
  color: #2c3e50;
  margin-bottom: 10px;
  border-bottom: 2px solid #3498db;
  padding-bottom: 5px;
}

/* Categories list */
.categories ul {
```

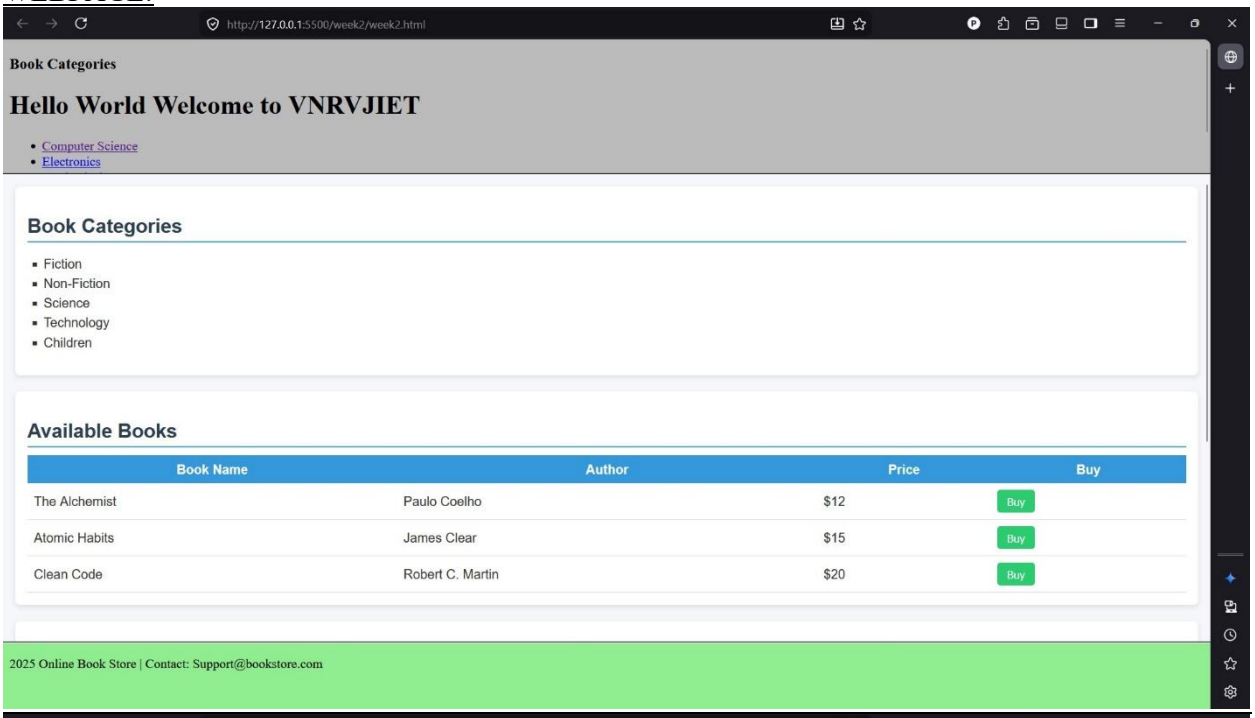
```
    list-style-type: square;
    padding-left: 20px;
}

.categories li {
    margin: 5px 0;
}
/* Table styling */
table {
    width: 100%;
    border-collapse: collapse;
}

th {
    background-color: #3498db;
    color: white;
    padding: 8px;
}

td {
    padding: 8px;
    border-bottom: 1px solid #ddd;
}

/* Buttons */
button {
    background-color: #2ecc71;
    color: white;
    border: none;
    padding: 6px 12px;
    border-radius: 4px;
    cursor: pointer;
}

button:hover {
    background-color: #27ae60;
}

/* Form styling */
form {
    display: grid;
    row-gap: 8px;
}

input, select {
    padding: 6px;
    border: 1px solid #ccc;
    border-radius: 4px;
}
```

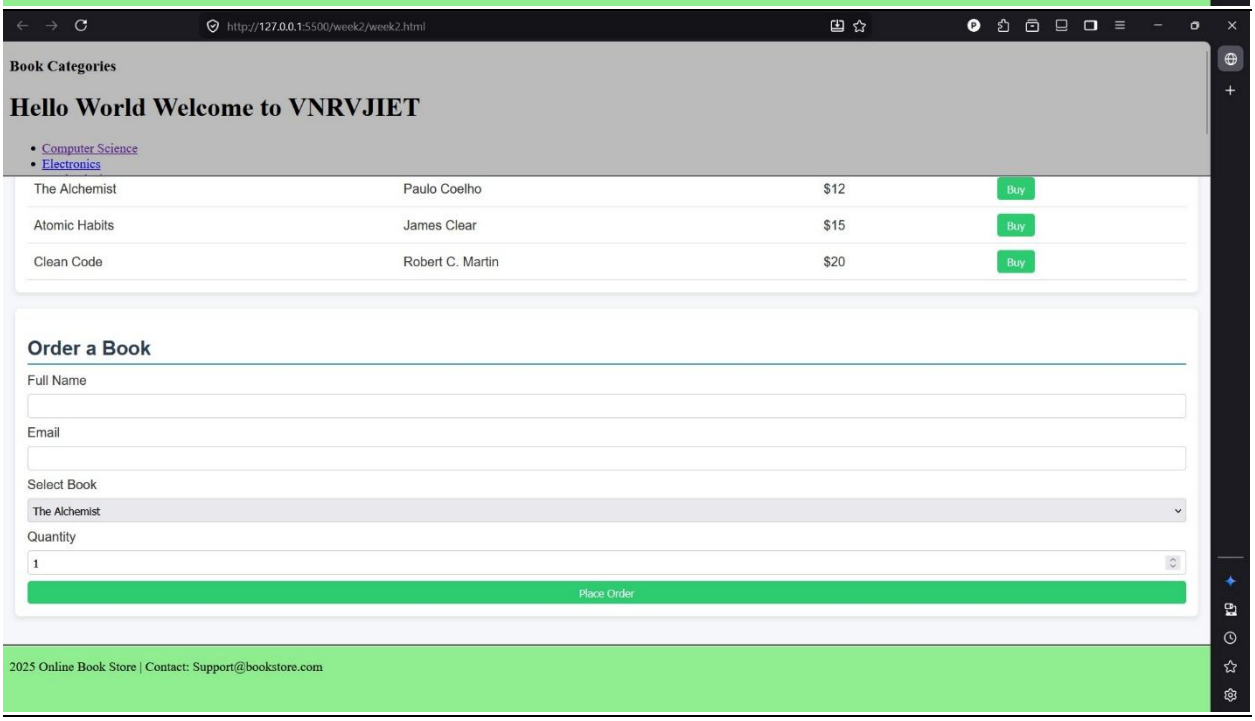
VNRVJIET

NAME OF LABORATORY: Web Technologies LAB

DATE: **EXERCISE NO:** **EXERCISE NAME:**

WEBPAGE:





WEEK-3:

JavaScript:

a) User Registration Form

Create a registration form for users signing up on the online bookstore. Validate the following:

- **Username: Should not be empty and must be at least 5 characters long.**
- **Password: Should not be empty and must be at least 8 characters long.**
- **Confirm Password: Should match the entered password.**

b) Forgot Password Form

Implement a form for users to reset their password. Validate the following:

- **Email Address: Should be a valid email format.**
- **Provide feedback if the email address is not associated with any account.**

c) Dynamically Content Loading

Use JavaScript to dynamically load book details into the content frame when a user clicks on a book in the navigation frame.

Week3.html:

```
<!DOCTYPE html>
<html>
   <head>
      <title>Online Book Store - Week 3</title>
   </head>
   <frameset rows="20%,70%,10%" border="2">
      <frame src="navigation.html" name="navFrame">
      <frame src="content.html" name="contentFrame">
      <frame src="footer.html" name="footerFrame">
   </frameset>
</html>
```

Week3.js:

```
// week3.js - Dynamically load book details into the content frame

document.addEventListener('DOMContentLoaded', function () {
   // This script runs inside the navigation frame (navigation.html)
   var navLinks = document.querySelectorAll('a[data-book-id]');
   if (!navLinks.length) {
      return;
   }

   var bookData = {
      'clean-code': {
         title: 'Clean Code',
         author: 'Robert C. Martin',
         category: 'Technology / Programming',
         price: '$20',
         isbn: '978-0132350884',
```

```
        availability: 'In Stock',
        description: 'A handbook of agile software craftsmanship focusing on writing clean, maintainable code.'
    },
    'alchemist': {
        title: 'The Alchemist',
        author: 'Paulo Coelho',
        category: 'Fiction',
        price: '$12',
        isbn: '978-0061122415',
        availability: 'In Stock',
        description: 'A novel about a shepherd boy on a journey to realize his personal legend.'
    },
    'atomic-habits': {
        title: 'Atomic Habits',
        author: 'James Clear',
        category: 'Self Help',
        price: '$15',
        isbn: '978-0735211292',
        availability: 'Limited Stock',
        description: 'Practical strategies for building good habits and breaking bad ones.'
    },
    'ds-algo': {
        title: 'Data Structures and Algorithms',
        author: 'Narasimha Karumanchi',
        category: 'Computer Science',
        price: '$25',
        isbn: '978-8193245279',
        availability: 'Out of Stock',
        description: 'Detailed explanations of core data structures and algorithms with interview focus.'
    }
};

function loadBookIntoContentFrame(bookId) {
    var info = bookData[bookId];
    if (!info) {
        return;
    }

    var contentFrame = parent.frames['contentFrame'];
    if (!contentFrame || !contentFrame.document) {
        return;
    }
```

```
        var doc = contentFrame.document;
        var container = doc.getElementById('bookDetails');
        if (!container) {
            return;
        }

        container.innerHTML =
            '<h3>' + info.title + '</h3>' +
            '<p><strong>Author:</strong> ' + info.author + '</p>' +
            '<p><strong>Category:</strong> ' + info.category + '</p>' +
            '<p><strong>ISBN:</strong> ' + info.isbn + '</p>' +
            '<p><strong>Price:</strong> ' + info.price + '</p>' +
            '<p><strong>Availability:</strong> ' + info.availability + '</p>' +
            '<p><strong>Description:</strong> ' + info.description + '</p>';
    }

    navLinks.forEach(function (link) {
        link.addEventListener('click', function (e) {
            e.preventDefault();
            var bookId = this.getAttribute('data-book-id');
            loadBookIntoContentFrame(bookId);
        });
    });
});
```

Registration.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>User Registration - Online Bookstore</title>
    <link rel="stylesheet" href="registration.css">
</head>
<body>
    <div class="container">
        <h2>User Registration</h2>
        <form id="registrationForm" autocomplete="off">
            <label for="username">Username:</label>
            <input type="text" id="username" name="username" required>

            <label for="password">Password:</label>
            <input type="password" id="password" name="password" required>

            <label for="confirmPassword">Confirm Password:</label>
            <input type="password" id="confirmPassword" name="confirmPassword" required>
```

```
        <div id="errorMsg" class="error" style="display:none;"></div>
        <button type="submit">Register</button>
        <div style="text-align:center; margin-top:15px;">
          <a href="forgotpassword.html">Forgot Password?</a>
        </div>
      </form>
    </div>
    <script src="registration.js"></script>
</body>
</html>
```

ForgotPassword.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Forgot Password - Online Bookstore</title>
    <link rel="stylesheet" href="forgotpassword.css">
</head>
<body>
    <div class="container">
      <h2>Forgot Password</h2>
      <form id="forgotPasswordForm" autocomplete="off">
        <label for="email">Email Address:</label>
        <input type="email" id="email" name="email" required>
        <div id="fpErrorMsg" class="error" style="display:none;"></div>
        <button type="submit">Reset Password</button>
      </form>
    </div>
    <script src="forgotpassword.js"></script>
</body>
</html>
```

Forgotpassword.js:

```
document.addEventListener('DOMContentLoaded', function() {
    var form = document.getElementById('forgotPasswordForm');
    if (!form) return;
    form.addEventListener('submit', function(e) {
      e.preventDefault();
      var email = document.getElementById('email').value.trim();
      var errorMsg = document.getElementById('fpErrorMsg');
      errorMsg.style.display = 'none';
      errorMsg.innerText = '';
```

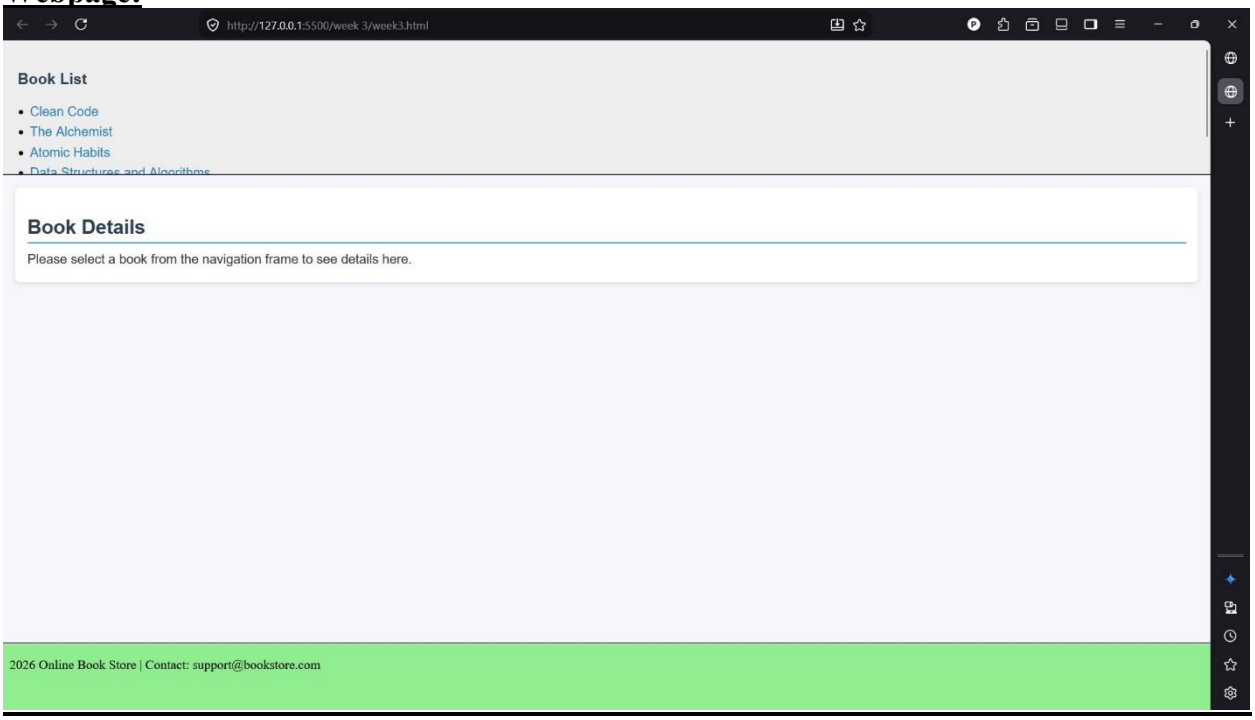
```
        var emailRegex = /^[\w-\.]+@([\w-]+\.)+[\w-]{2,4}$/;
        if (!emailRegex.test(email)) {
            errorMsg.innerText = 'Please enter a valid email address.';
            errorMsg.style.display = 'block';
            return;
        }

        // Simulated check for email association (for demo, only these emails are valid)
        var validEmails = ['user1@example.com', 'user2@example.com',
'test@bookstore.com','pranav.tungani@gmail.com'];
        if (!validEmails.includes(email)) {
            errorMsg.innerText = 'This email address is not associated with any account.';
            errorMsg.style.display = 'block';
            return;
        }

        alert('A password reset link has been sent to your email.');
        form.reset();
    });
});
```

Registration.js:

```
// registration.js - Security and validation for registration form

document.addEventListener('DOMContentLoaded', function() {
    var form = document.getElementById('registrationForm');
    if (!form) return;
    form.addEventListener('submit', function(e) {
        e.preventDefault();
        var username = document.getElementById('username').value.trim();
        var password = document.getElementById('password').value;
        var confirmPassword = document.getElementById('confirmPassword').value;
        var errorMsg = document.getElementById('errorMsg');
        errorMsg.style.display = 'none';
        errorMsg.innerText = '';

        // Username validation
        if (username.length < 5) {
            errorMsg.innerText = 'Username must be at least 5 characters long.';
            errorMsg.style.display = 'block';
            return;
        }
        // Password validation
        if (password.length < 8) {
            errorMsg.innerText = 'Password must be at least 8 characters long.';
            errorMsg.style.display = 'block';
```

```
        return;
    }
    // Password strength (at least 1 uppercase, 1 lowercase, 1 number)
    var strongRegex = /^(?=.*[a-z])(?=.*[A-Z])(?=.*\d).+$/;
    if (!strongRegex.test(password)) {
        errorMsg.innerText = 'Password must contain uppercase, lowercase, and a number.';
        errorMsg.style.display = 'block';
        return;
    }
    // Confirm password
    if (password !== confirmPassword) {
        errorMsg.innerText = 'Passwords do not match.';
        errorMsg.style.display = 'block';
        return;
    }
    // All validations passed
    alert('Registration successful!');
    form.reset();

    // Navigate to the content page after successful registration
    window.location.href = 'navigation.html';
  });
});
```

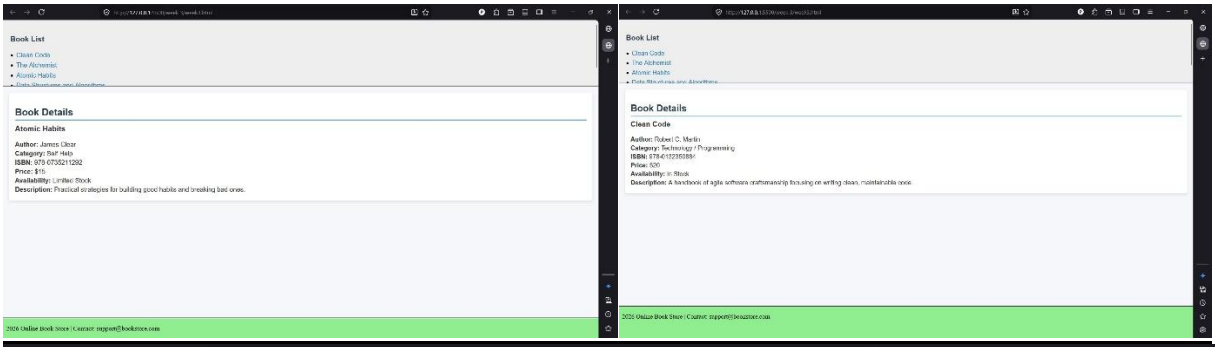
Webpage:

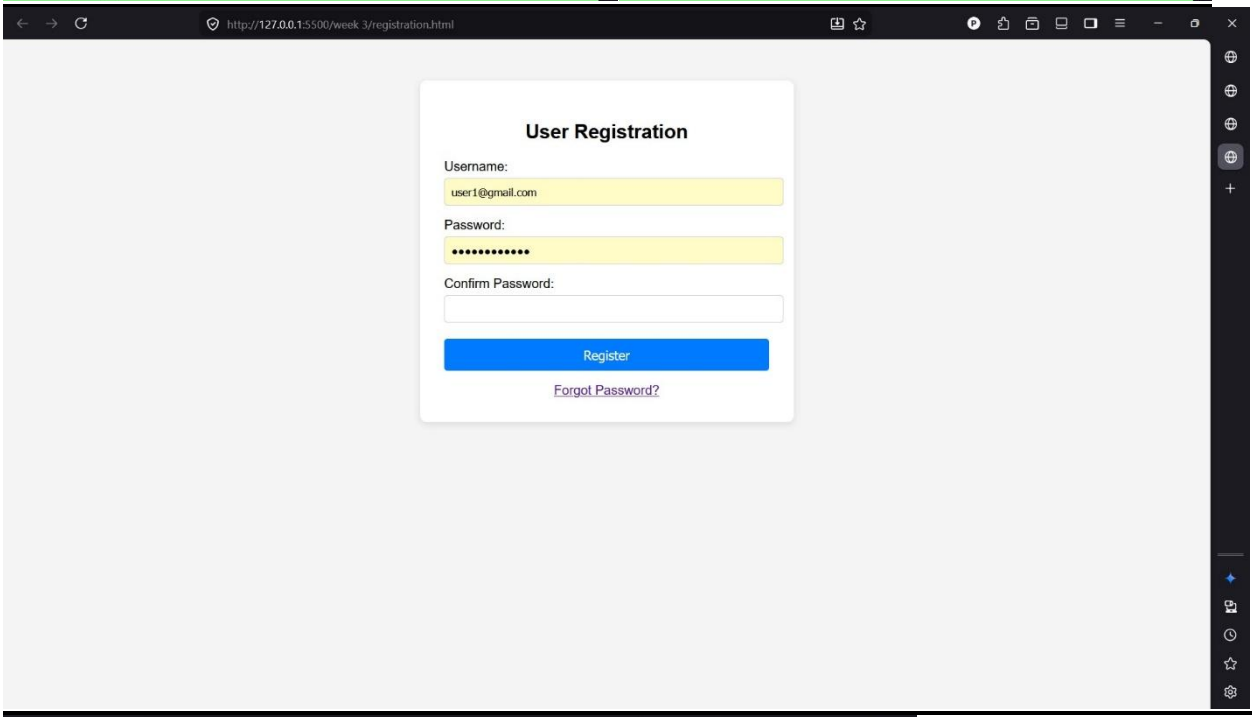


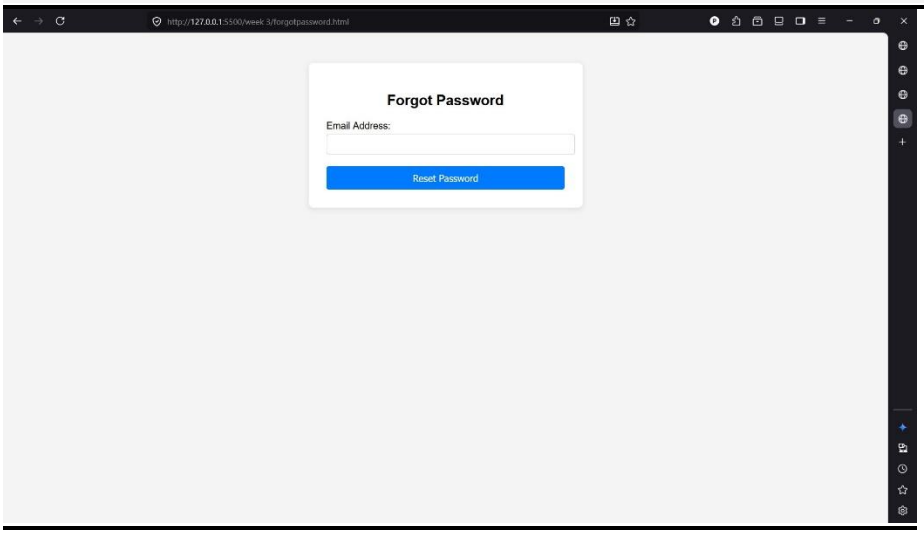
VNRVJIET

NAME OF LABORATORY: Web Technologies LAB

DATE: **EXERCISE NO:** **EXERCISE NAME:**







DATE: **EXERCISE NO:** **EXERCISE NAME:**

WEEK-4: Shopping Cart Functionality: Implement a simple shopping cart using JavaScript. Allow users to add books to their cart and view the cart's contents.

Cart.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>My Cart</title>
   <link rel="stylesheet" href="styles.css">
</head>
<body>
   <main>
   <h1>My Cart</h1>
   <a href="index.html">Back to Store</a>

   <div id="cart-items"></div>
   <p><strong>Total: Rs. <span id="total">0</span></strong></p>
   <button onclick="clearCart()">Clear Cart</button>
   </main>

   <script src="script.js"></script>
</body>
</html>
```

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Book Store</title>
   <link rel="stylesheet" href="styles.css">
</head>
<body>
   <main>
   <h1>Book Store</h1>
   <a href="cart.html">View Cart (<span id="cart-count">0</span>)</a>

   <div id="books">
      <div class="book">
      <h3>The Alchemist</h3>
      <p>By Paulo Coelho</p>
      <p>Price: Rs. 299</p>
      <button onclick="addToCart('The Alchemist', 299)">Add to Cart</button>
   </div>
```

```
    <div class="book">
        <h3>Atomic Habits</h3>

        <p>By James Clear</p>
        <p>Price: Rs. 499</p>
        <button onclick="addToCart('Atomic Habits', 499)">Add to Cart</button>
    </div>

    <div class="book">
        <h3>Rich Dad Poor Dad</h3>
        <p>By Robert Kiyosaki</p>
        <p>Price: Rs. 349</p>
        <button onclick="addToCart('Rich Dad Poor Dad', 349)">Add to Cart</button>
    </div>

    <div class="book">
        <h3>Deep Work</h3>
        <p>By Cal Newport</p>
        <p>Price: Rs. 399</p>
        <button onclick="addToCart('Deep Work', 399)">Add to Cart</button>
    </div>
    </div>
    </main>

    <script src="script.js"></script>
</body>
</html>
```

Script.js:

```
// Get cart from localStorage (or empty array)
function getCart() {
    return JSON.parse(localStorage.getItem('cart')) || [];
}

// Save cart to localStorage
function saveCart(cart) {
    localStorage.setItem('cart', JSON.stringify(cart));
}

// Add a book to the cart
function addToCart(title, price) {
    var cart = getCart();
    cart.push({ title: title, price: price });
    saveCart(cart);
    alert(title + ' added to cart!');
    updateCount();
```

```
}

// Update the cart count shown on the store page
function updateCount() {
    var countEl = document.getElementById('cart-count');
    if (countEl) {
        countEl.textContent = getCart().length;
    }
}

// Show cart items on cart.html
function showCart() {
    var cart = getCart();
    var container = document.getElementById('cart-items');
    if (!container) return;

    if (cart.length === 0) {
        container.innerHTML = '<p>Your cart is empty.</p>';
        document.getElementById('total').textContent = 0;
        return;
    }

    var html = '<ul>';
    var total = 0;
    for (var i = 0; i < cart.length; i++) {
        html += '<li>' + cart[i].title + ' - Rs. ' + cart[i].price + '</li>';
        total += cart[i].price;
    }
    html += '</ul>';
    container.innerHTML = html;
    document.getElementById('total').textContent = total;
}

// Clear all items from cart
function clearCart() {
    localStorage.removeItem('cart');
    showCart();
}

// Run on page load
updateCount();
showCart();
```

Styles.css:

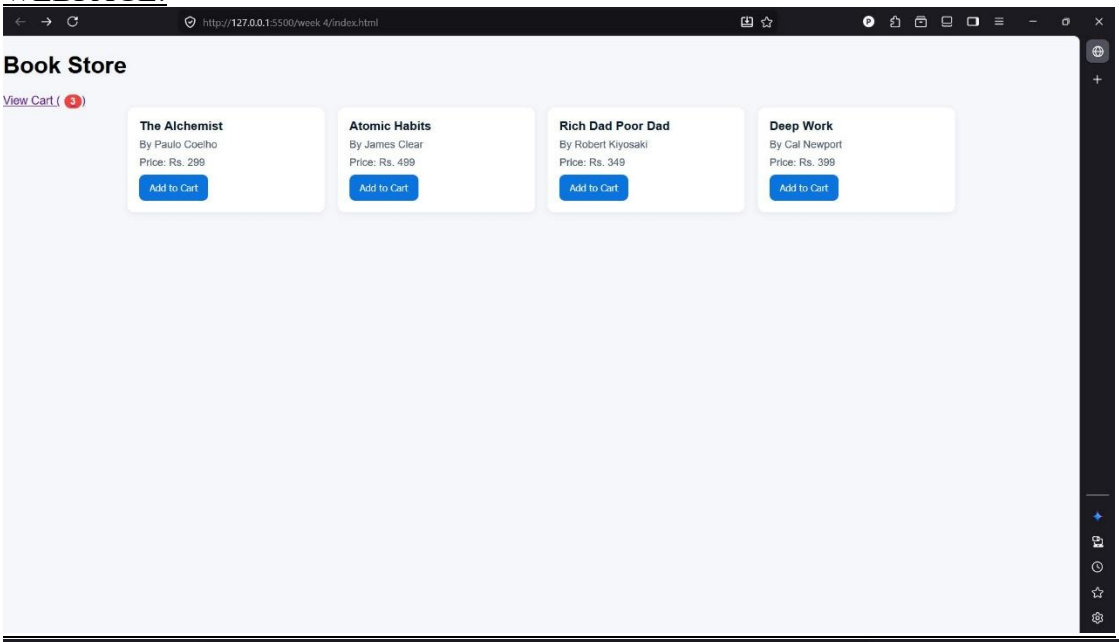
```
body {
    font-family: Arial, sans-serif;
```

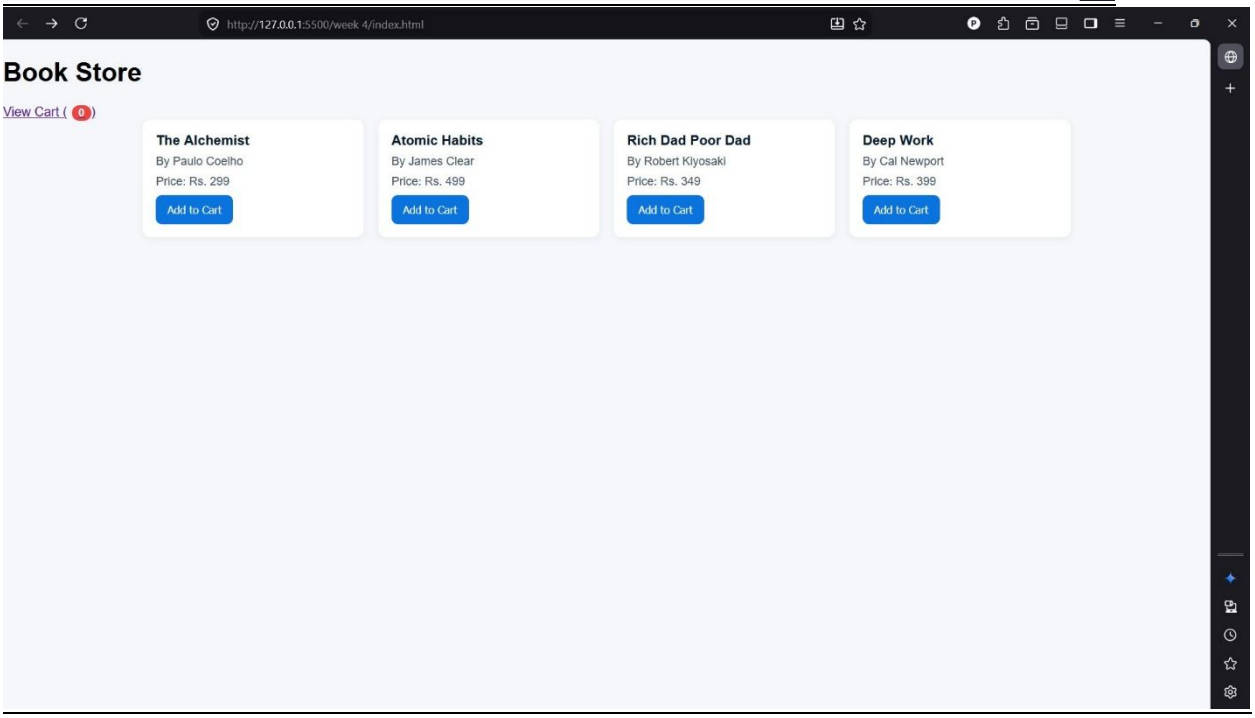
```
    margin: 0;
    padding: 0;
    background: #f5f7fb;
}
/* Books grid */
#books{
    display: flex;
    flex-wrap: wrap;
    gap: 18px;
    justify-content: center;
}

.book {
    background: #ffffff;
    border-radius: 10px;
    box-shadow: 0 2px 8px rgba(11,37,69,0.06);
    padding: 16px;
    width: 240px;
    display: flex;
    flex-direction: column;
    gap: 8px;
    transition: transform 160ms ease, box-shadow 160ms ease;
}

.book:hover {
    transform: translateY(-6px);
    box-shadow: 0 10px 22px rgba(11,37,69,0.10);
}

.book h3 { margin: 0; font-size: 1.05rem; color: #071b2d; }
.book p { margin: 0; color: #475569; font-size: 0.9rem; }
.book .price { font-weight: 700; color: #111827; }

button {
    padding: 10px 14px;
    background-color: #0b74de;
    color: white;
    border: none;
    border-radius: 8px;
    cursor: pointer;
    align-self: start;
    transition: background-color 120ms ease, transform 120ms ease;
}

button:hover { background-color: #075fae; transform: translateY(-2px); }
```

```
button:active { transform: translateY(0); }

/* Cart link badge */
#cart-count {
   background: #ef4444;
   color: #fff;
   padding: 2px 8px;
   border-radius: 999px;
   margin-left: 6px;
   font-weight: 700;
   font-size: 0.85rem;
}

/* Cart page styles */
#cart-items { margin: 18px 0; }
.cart-item {
   display: flex;
   justify-content: space-between;
   align-items: center;
   padding: 10px 12px;
   background: #fff;
   border-radius: 8px;
   box-shadow: 0 1px 4px rgba(11,37,69,0.04);
   margin-bottom: 10px;
}

.cart-item .name { font-weight: 600; }
.cart-item .price { color: #0b2545; font-weight: 700; }

/* Responsive */
@media (max-width: 600px) {
   main { margin: 18px; padding: 12px; }
   .book { width: 100%; max-width: 420px; }
}

/* Small utility */
.muted { color: #6b7280; font-size: 0.9rem; }

/* Style cart list rendered by script.js */
#cart-items ul { list-style: none; padding: 0; margin: 0; }
#cart-items ul li {
   padding: 10px 12px;
   background: #fff;
   border-radius: 8px;
   box-shadow: 0 1px 4px rgba(11,37,69,0.04);
```

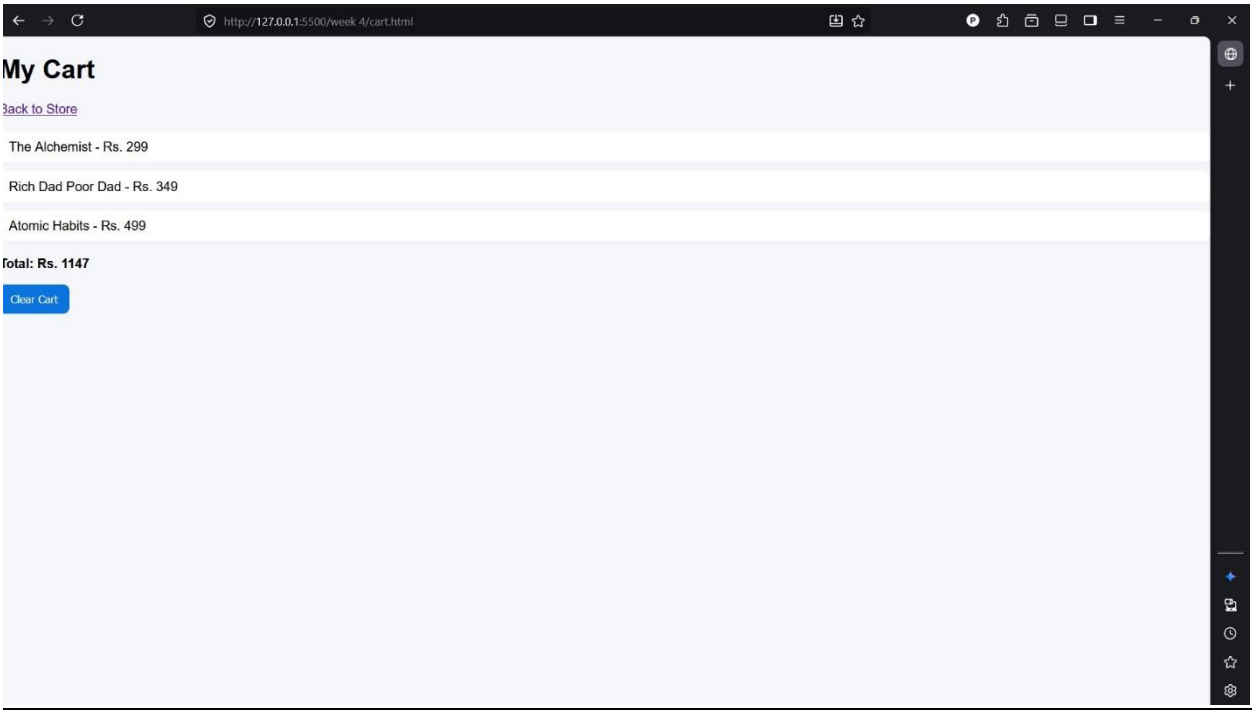
```
    margin-bottom: 10px;
    display: flex;
    justify-content: space-between;
    align-items: center;
}
#cart-items ul li::after { content: attr(data-price); font-weight:700; color:#0b2545; }
```

WEBPAGE:





DATE: **EXERCISE NO:** **EXERCISE NAME:**



DATE: **EXERCISE NO:** **EXERCISE NAME:**

WEEK-5: XML Structure and Design: Book Catalog XML: Design an XML structure to represent the catalog of books in the online store. Include details such as book title, author, genre, price, and availability.

Book_catalog_view.html:

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width,initial-scale=1">
  <title>Book Catalog Viewer</title>
  <link rel="stylesheet" href="book_catalog.css">
</head>
<body>
  <header class="site-header">
    <h1>Book Catalog</h1>
  </header>

  <main class="container">
    <div class="controls">
      <input id="search" placeholder="Search by title, author, or genre" />
      <select id="genreFilter"><option value="">All genres</option></select>
      <button id="refresh">Refresh</button>
    </div>

    <div id="catalog" class="grid" aria-live="polite"></div>
  </main>

  <script src="book_catalog.js"></script>
</body>
</html>
```

Book_catalog.css:

```
:root{
  --bg:#f6f8fa;
  --card:#ffffff;
  --muted:#6b7280;
  --accent:#2563eb;
  --success:#10b981;
  --danger:#ef4444;
  --glass:rgba(255,255,255,0.6);
}
*{box-sizing:border-box}
body{font-family:Segoe UI,Roboto,Arial,sans-serif;margin:0;background:var(--bg);color:#111}
.site-header{background:linear-gradient(90deg,var(--accent),#7c3aed);color:white;padding:20px
24px}
```

```
.site-header h1{margin:0;font-size:1.25rem}
.container{max-width:1100px;margin:24px auto;padding:0 16px}
.controls{display:flex;gap:10px;align-items:center;margin-bottom:18px}
.controls input, .controls select{padding:10px 12px;font-size:0.95rem;border:1px solid #e6e9ee;border-radius:8px;min-width:0}
.controls button{padding:10px 14px;border-radius:8px;background:var(--accent);color:white;border:none;cursor:pointer}
.grid{display:grid;grid-template-columns:repeat(auto-fill,minmax(220px,1fr));gap:16px}
.card{background:var(--card);padding:14px;border-radius:12px;box-shadow:0 6px 20px rgba(20,20,40,0.06);display:flex;flex-direction:column;gap:8px}
.title{font-weight:700;font-size:1.02rem}
.author{color:var(--muted);font-size:0.92rem}
.meta{display:flex;justify-content:space-between;align-items:center;margin-top:6px}
.genre{font-size:0.85rem;padding:6px 8px;border-radius:999px;background:var(--glass);color:#374151}
.price{font-weight:700;color:var(--accent)}
.availability{font-size:0.85rem;padding:6px 8px;border-radius:999px}
.availability.in-stock{background:rgba(16,185,129,0.08);color:var(--success);border:1px solid rgba(16,185,129,0.12)}
.availability.out-of-stock{background:rgba(239,68,68,0.08);color:var(--danger);border:1px solid rgba(239,68,68,0.12)}
.empty{padding:30px;text-align:center;color:var(--muted)}
@media (max-width:520px){.controls{flex-direction:column;align-items:stretch}.controls input{width:100%}}
```

Book_catalog.js:

```
async function loadCatalog(){
  try{
    const res = await fetch('book_catalog.xml');
    if(!res.ok) throw new Error('Failed to load XML');
    const text = await res.text();
    const parser = new DOMParser();
    const xml = parser.parseFromString(text, 'application/xml');

    const books = Array.from(xml.getElementsByTagName('book')).map(b => {
      const title = b.getElementsByTagName('title')[0]?.textContent || '';
      const author = b.getElementsByTagName('author')[0]?.textContent || '';
      const genre = b.getElementsByTagName('genre')[0]?.textContent || '';
      const priceEl = b.getElementsByTagName('price')[0];
      const price = priceEl?.textContent || '';
      const currency = priceEl?.getAttribute('currency') || '';
      const availEl = b.getElementsByTagName('availability')[0];
      const availabilityCount = availEl?.textContent || '0';
      const status = availEl?.getAttribute('status') || (Number(availabilityCount)>0 ? 'in-stock' : 'out-of-stock');
```

```
        return { id: b.getAttribute('id'), title, author, genre, price, currency, availabilityCount: Number(availabilityCount), status };
    });

    return books;
  }catch(err){
    console.error(err);
    return null;
  }
}

function renderBooks(books){
  const container = document.getElementById('catalog');
  container.innerHTML = '';
  if(!books || books.length===0){
    container.innerHTML = '<div class="empty">No books found.</div>';
    return;
  }

  const frag = document.createDocumentFragment();
  books.forEach(b=>{
    const card = document.createElement('article');
    card.className = 'card';
    card.innerHTML = `
      <div class="title">${escapeHtml(b.title)}</div>
      <div class="author">by ${escapeHtml(b.author)}</div>
      <div class="meta">
        <div class="genre">${escapeHtml(b.genre)}</div>
        <div class="price">${escapeHtml(b.currency)} ${escapeHtml(b.price)}</div>
      </div>
      <div style="display:flex;justify-content:space-between;align-items:center">
        <div class="availability ${b.status === 'in-stock' ? 'in-stock' : 'out-of-stock'}">${b.status === 'in-stock' ? 'In stock' : 'Out of stock'}</div>
        <div style="color:var(--muted);font-size:0.85rem">Qty: ${b.availabilityCount}</div>
      </div>
    `;
    frag.appendChild(card);
  });
  container.appendChild(frag);
}

function escapeHtml(s){
  return String(s).replaceAll('&','&amp;').replaceAll('<','&lt;').replaceAll('>','&gt;');
}
```

```
function applyFilters(allBooks){
  const q = document.getElementById('search').value.trim().toLowerCase();
  const genre = document.getElementById('genreFilter').value;
  let filtered = allBooks.filter(b => {
    const hay = `${b.title} ${b.author} ${b.genre}`.toLowerCase();
    return hay.includes(q) && (genre === '' || b.genre === genre);
  });
  renderBooks(filtered);
}

(async function init(){
  const allBooks = await loadCatalog();
  if(allBooks === null){
      document.getElementById('catalog').innerHTML = '<div class="empty">Could not load catalog. Make sure <code>book_catalog.xml</code> is next to the viewer.</div>';
    return;
  }

  // populate genre filter
  const genres = Array.from(new Set(allBooks.map(b=>b.genre).filter(Boolean))).sort();
  const select = document.getElementById('genreFilter');
  genres.forEach(g=>{
    const opt = document.createElement('option');
    opt.value = g; opt.textContent = g; select.appendChild(opt);
  });

  // initial render
  renderBooks(allBooks);

  // wire controls
  document.getElementById('search').addEventListener('input', ()=>applyFilters(allBooks));
  select.addEventListener('change', ()=>applyFilters(allBooks));
  document.getElementById('refresh').addEventListener('click', async ()=>{
    const refreshed = await loadCatalog();
    if(refreshed) {
      allBooks.length = 0; allBooks.push(...refreshed);
      applyFilters(allBooks);
    }
  });
})();
```

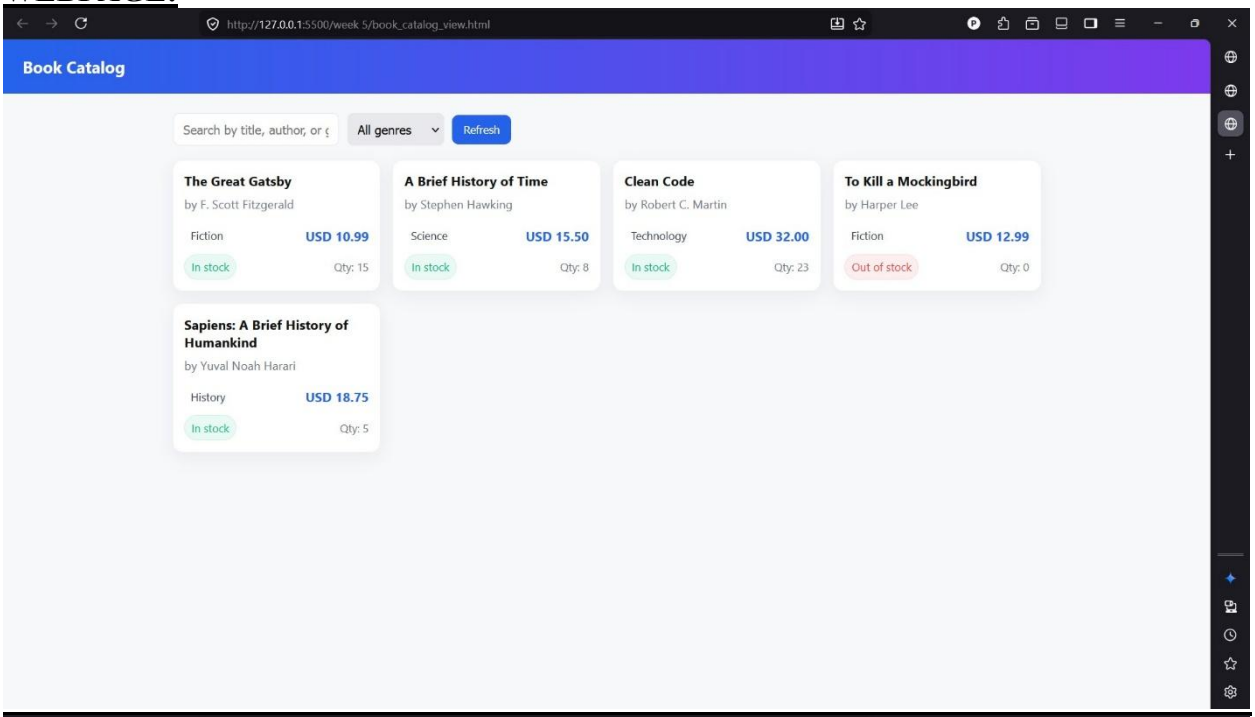
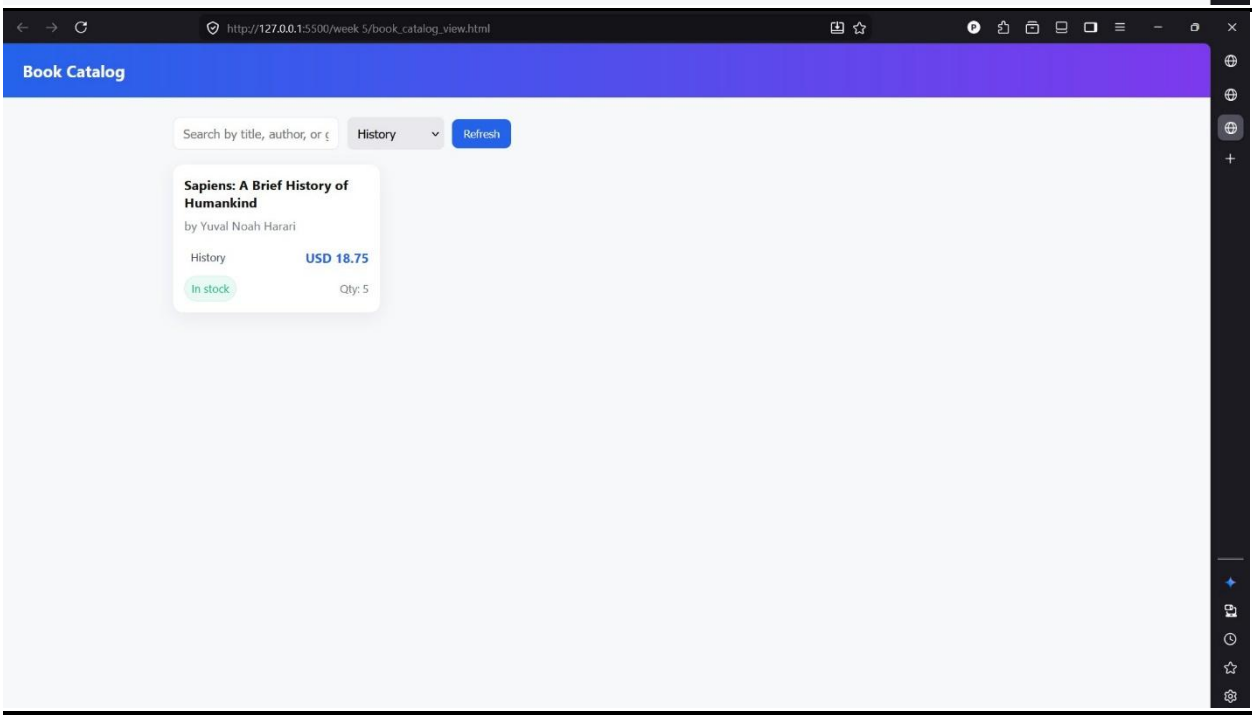
Book_catalog.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<catalog>
   <book id="B001">
```

```
    <title>The Great Gatsby</title>
    <author>F. Scott Fitzgerald</author>
    <genre>Fiction</genre>
    <price currency="USD">10.99</price>
    <availability status="in-stock">15</availability>
  </book>

  <book id="B002">
    <title>A Brief History of Time</title>
    <author>Stephen Hawking</author>
    <genre>Science</genre>
    <price currency="USD">15.50</price>
    <availability status="in-stock">8</availability>
  </book>

  <book id="B003">
    <title>Clean Code</title>
    <author>Robert C. Martin</author>
    <genre>Technology</genre>
    <price currency="USD">32.00</price>
    <availability status="in-stock">23</availability>
  </book>

  <book id="B004">
    <title>To Kill a Mockingbird</title>
    <author>Harper Lee</author>
    <genre>Fiction</genre>
    <price currency="USD">12.99</price>
    <availability status="out-of-stock">0</availability>
  </book>

  <book id="B005">
    <title>Sapiens: A Brief History of Humankind</title>
    <author>Yuval Noah Harari</author>
    <genre>History</genre>
    <price currency="USD">18.75</price>
    <availability status="in-stock">5
  </book>
</catalog>
```

VNRVJIET

NAME OF LABORATORY: Web Technologies LAB

DATE: **EXERCISE NO:** **EXERCISE NAME:**

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